

-----AJAY BHOSALE-----

\* Q1. Write a Function to classify a person's age group: Child, Teen, Adult, senior.

```
def age_group(age):  
    if age>=0 and age<=3:  
        print(f"Age Group: Baby {age}yr")  
    elif age>=4 and age<=12:  
        print(f"Age Group: Child {age}yrs")  
    elif age>=13 and age<=19:  
        print(f"Age Group: Teen {age}yrs")  
    elif age>=20 and age<=59:  
        print(f"Age Group: Adult {age}yrs")  
    elif age>59:  
        print(f"Age Group: Senior Citizen {age}yrs")  
age1 = int(input("Enter your age: "))  
age_group(age1)
```

Enter your age: 8  
Age Group: Child 8yrs

\* Q2. Write a Function to Check is string is palindrome.

```
def word_palindrome(word):  
    revWord = ""  
    for j in word:  
        i = j.lower()  
        revWord = i + revWord  
    print("Reverse Word:",revWord)  
    if word.lower() == revWord:  
        print("Palindrome")  
    else:  
        print("Not Palindorme")
```

```
words = input("Enter any words: ")  
word_palindrome(words)
```

Enter any words: Racecar  
Reverse Word: racecar  
Palindrome

\* Q3. Write a function to check if a number is within a given range (1-100).

```
def num_range(num):
```

```
    if num >= 1 and num <= 100:
```

```
        print(f"The entered number {num} is within range(1-100)")
```

```
    else:
```

```
        print("Please enter number in range(1-100)")
```

```
num1 = int(input("Enter a number: "))
```

```
num_range(num1)
```

Enter a number: 75

The entered number 75 is within range(1-100)

\* Q4. Write a function to find the sum of numbers from 1 to N.

```
def sum_numbers(n):
```

```
    i = 1
```

```
    sum1 = 0
```

```
    for i in range(1, n+1):
```

```
        sum1 = sum1 + i
```

```
    else:
```

```
        print(f"Sum of the value of given number {n} is {sum1}")
```

```
N = int(input("Enter value for 'N': "))
```

```
sum_numbers(N)
```

Enter value for 'N': 11

Sum of the value of given number 11 is 66

\* Q5. Write a function to take N users number and find the factorial of all numbers.

```
def fact_num(N):  
    i = 1  
    fact = 1  
  
    for i in range(1,N+1):  
        fact = fact*i  
  
    else:  
        print(f"Factorial of N users number {N} is {fact}")  
N1 = int(input("Enter value for 'N': "))  
fact_num(N1)
```

Enter value for 'N': 5  
Factorial of N users number 5 is 120

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\* Q6. Write a function to print all numbers divisible by 7 but not by 5 from 1-100.

```
def num_div():  
    for i in range(1,100):  
        if i%7==0 and i%5!=0:  
            print(i,end= " ")  
num_div()
```

7 14 21 28 42 49 56 63 77 84 91 98

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\* Q9. Write a function to take N user numbers and display even and odd separately and find their sum and average.

```
def user_num(N):
```

```
    i = 1
```

```
    sum1 = 0
```

```
    sum2 = 0
```

```
    count1 = 1
```

```
    count2 = 1
```

```
    for i in range(i,N+1):
```

```
        if i%2==0 and count1<=N:
```

```
            sum1 = sum1 + i
```

```
            count1+=1
```

```
            avg1 = sum1/count1
```

```
        for i in range(i,N+1):
```

```
            if i%2!=0 and count2<=N:
```

```
                sum2 = sum2 + i
```

```
                count2+=1
```

```
                avg2 = sum2/count2
```

```
    else:
```

```
        print(f"Sum of the value of given number {N} for even number is {sum1}")
```

```
        print(f"Average of the value of given number {N} for even number is {avg1}")
```

```
        print(f"Sum of the value of given number {N} for odd number is {sum2}")
```

```
        print(f"Average of the value of given number {N} for odd number is {avg2}")
```

```
N1 = int(input("Enter value for 'N': "))
```

```
user_num(N1)
```

Enter value for 'N': 20

Sum of the value of given number 20 for even number is 110

Average of the value of given number 20 for even number is 10.0

Sum of the value of given number 20 for odd number is 202

Average of the value of given number 20 for odd number is 9.619047619047619

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\* Q10. Create a function to count the number of vowels and consonants in a string.  
String = "The quick brown fox jumps over the lazy dog"

vowels = "aeiou"

string = "The quick brown fox jumps over the lazy dog"

def vow\_cons():

i = 0

vowChar = []

vowCount = 0

consChar = []

consCount = 0

while i < len(string)-1:

print(i, string[i])

i += 1

if string[i].lower() in vowels:

vowChar.append(string[i])

vowCount += 1

else:

if string[i] == " ":

continue

consChar.append(string[i])

consCount += 1

print("Vowel characters list:", vowChar, "=", vowCount)

print("Consonent characters list:", consChar, "=", consCount)

vow\_cons()

Vowel characters list: ['e', 'u', 'i', 'o', 'o', 'u', 'o', 'e', 'e', 'a', 'o'] = 11

Consonent characters list: ['h', 'q', 'c', 'k', 'b', 'r', 'w', 'n', 'f', 'x', 'j', 'm', 'p', 's', 'v', 'r', 't', 'h', 'l', 'z', 'y', 'd', 'g'] = 23

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\* Q14. Write a function to display cube of numbers from 1-50 in a list.

```
def cube_numbers():  
    return [num**3 for num in range(1, 51)]
```

```
words = ["Tranquil", "Meadow", "Illuminate", "Whisper", "Serene", "Cascade", "Ember", "Twilight",  
"Glimmer", "Solitude", "Wander", "Horizon", "Luminous", "Echo", "Drift", "Radiant", "Murmur",  
"Velvet", "Breeze", "Ethereal"]  
print("Words:", words)  
print(cube_numbers())
```

```
[1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, 1331, 1728, 2197, 2744, 3375, 4096, 4913,  
5832, 6859, 8000, 9261, 10648, 12167, 13824, 15625, 17576, 19683, 21952, 24389,  
27000, 29791, 32768, 35937, 39304, 42875, 46656, 50653, 54872, 59319, 64000,  
68921, 74088, 79507, 85184, 91125, 97336, 103823, 110592, 117649, 125000]
```

Q.19 What is polymorphism? Explain in details.

Polymorphism is defined as;

A child class redefines a method that is already exists in parent class.

Polymorphism using One Function for Different Object (Duck Typing)

Method Overriding in Python is where a child class provides its own implementation of a method that is already defined in the parent class.

Selection deleted

#Example:

```
class Animal:
```

```
    def sound(self):
```

```
        print("Animal Makes a sound")
```

```
class Dog(Animal):
```

```
    def sound(self):
```

```
        print("Dog Barks") # overriding
```

```
class Cat(Animal):
```

```
    def sound(self):
```

```
        print("Mew") # overriding
```

```
d = Dog()
```

```
d.sound()
```

```
c = Cat()
```

```
c.sound()
```

Dog Barks

Mew

Polymorphism uses Operator Overloading:

Same operator behaves differently depending on datatype.

#example:

```
print(10+15) # calculation
```

```
print("Ajay "+"Bhosale") # concatenation
```

```
print([1,2]+["Hello", "World"]) # List merging
```

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Ajay Bhosale

[1, 2, 'Hello', 'World']